



MSME - HI AVNIET

MSME Idea Hackathon 6.0

TITLE : FactorySense contactless machine health monitoring system using WiFi Radar

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Introduction



Traditional machine monitoring requires physical sensors.



Installing sensors on every machine is **expensive** and **difficult**.



Unexpected machine failures cause production **downtime** and financial **losses**.



FactorySense uses existing WiFi signals to monitor machine health **without attaching sensors**.



Detects abnormal vibrations and movement for **early fault detection**.



Concept & Objectives



Concept

Our system uses WiFi signals as a radar to monitor machine health **without attaching any physical sensors**. By analyzing changes in the reflected WiFi signals, it can detect abnormal vibrations or movement and provide an early warning before a machine fails.



Objectives



Detect abnormal machine vibrations without physical sensors.



Monitor multiple machines using a single WiFi device.



Provide early fault alerts to reduce unexpected breakdowns.



Lower installation and maintenance costs.



Improve factory productivity and worker safety.

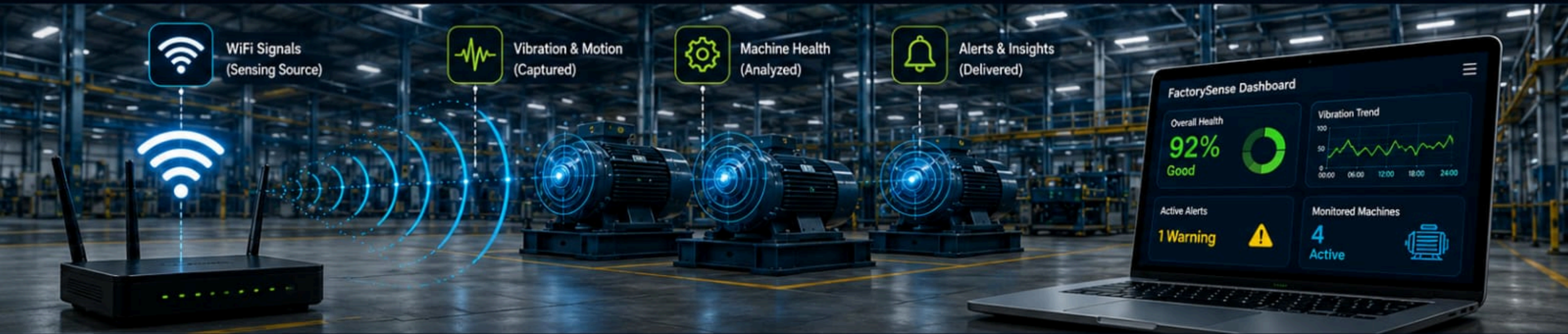
INNOVATION & UNIQUENESS

What makes **FactorySense** different?



FactorySense

Contactless Machine
Health Monitoring



KEY INNOVATIONS



CONTACTLESS MONITORING

No sensors are attached to the machine. The system uses existing WiFi signals to observe machine behavior.



ONE DEVICE, MULTIPLE MACHINES

A single WiFi access point can monitor several machines, reducing hardware requirements.



LOWER INSTALLATION COST

No wiring, sensor installation, or frequent sensor replacement is needed. Ideal for MSMEs.



EARLY FAULT DETECTION

Small changes in vibration or movement are identified before they become major machine failures.



CONTINUOUS REAL-TIME MONITORING

Machine health is monitored 24x7, providing real-time insights for quick maintenance action.



EASY TO SCALE

New machines can be added anytime without complex setup or system redesign.

HOW WE ARE DIFFERENT

FEATURE	TRADITIONAL METHODS	FACTORYSENSE
 SENSING METHOD	Physical sensors on every machine	Contactless WiFi sensing – no physical sensors
 COST	High installation and maintenance cost	Uses existing WiFi infrastructure – cost effective
 MONITORING CAPACITY	Monitors one machine at a time	One system monitors multiple machines
 MAINTENANCE	Regular sensor maintenance and calibration	Minimal maintenance and higher reliability

“

Turning everyday WiFi signals into a **smart industrial health monitoring** system. ”



Reduces
Downtime



Improves
Safety



Boosts
Productivity



Supports
MSME Growth

POTENTIAL AREAS OF APPLICATION

One system. Many industries. **Real impact.**



FactorySense

Contactless Machine
Health Monitoring



ROTATING MACHINERY

Motors, pumps, fans, compressors – detect abnormal vibrations early and prevent failures.



CONVEYOR SYSTEMS

Monitor belt movement and rollers for slippage, misalignment, or blockages.



CNC & MACHINE TOOLS

Track machine health and spindle condition to avoid precision loss and unexpected downtime.



ROBOTICS & AUTOMATION EQUIPMENT

Ensure smooth operation of robotic systems and automated production lines.



ELECTRICAL PANELS & SWITCHGEAR

Detect abnormal heating, loose connections, or vibrations in panels.



WAREHOUSES & STORAGE UNITS

Monitor large areas and equipment without installing any sensors.



CHEMICAL & PROCESS INDUSTRY

Monitor pumps, agitators, and compressors for safe and continuous operations.



FOOD & BEVERAGE INDUSTRY

Ensure smooth running of mixers, conveyors, and packaging machines.



From small workshops to large industries,
FactorySense fits everywhere.



Uses existing
WiFi
infrastructure



Monitors multiple
machines at once



Cost-effective
solution



Improves safety
& productivity



MARKET POTENTIAL OF OUR INNOVATION / IDEA

A large and growing market ready for **smarter, contactless** solutions.



FactorySense

Contactless Machine
Health Monitoring



The industrial IoT market in India is growing rapidly as industries shift towards **automation, real-time monitoring** and **predictive maintenance**.



IIoT Market in India
(2023)

₹ 15,230
CRORE+

The Indian IIoT market
is rapidly expanding.

Source: IMARC



IIoT Market in India
(2030)

₹ 48,555
CRORE+

Expected to grow at a
strong pace in the coming years.

Source: IMARC



CAGR
(2024–2030)

18.3%

High growth driven by
industry digitalization,
AI & IoT adoption.

Source: Fortune Business Insights

WHO CAN BENEFIT?



Manufacturing
Plants



Process
Industries



Heavy
Engineering
Units



Warehouses &
Logistics



Utilities &
Energy
Sector

WHY THE MARKET IS READY



Rising downtime
& maintenance cost

Unplanned downtime is
very expensive for
industries.



Need for cost-effective
solutions

MSMEs need affordable
monitoring solutions
without complex sensors.



Rise in IoT & Wi-Fi
infrastructure

Widespread availability
of Wi-Fi creates an
opportunity for
contactless monitoring.



Focus on Industry 4.0
& smart factories

Companies are investing in
digital solutions for better
efficiency and productivity.

OUR TARGET MARKET



Primary Focus
Small & Medium
Manufacturing Enterprises
(MSMEs) across India



Future Scope
Expansion to large scale
industries and global
markets



FactorySense taps into a high-growth market
with a **simple, scalable, and affordable** solution
that industries truly need.



Uses existing
Wi-Fi
infrastructure



Cost-effective
for MSMEs



Easy to deploy
& maintain

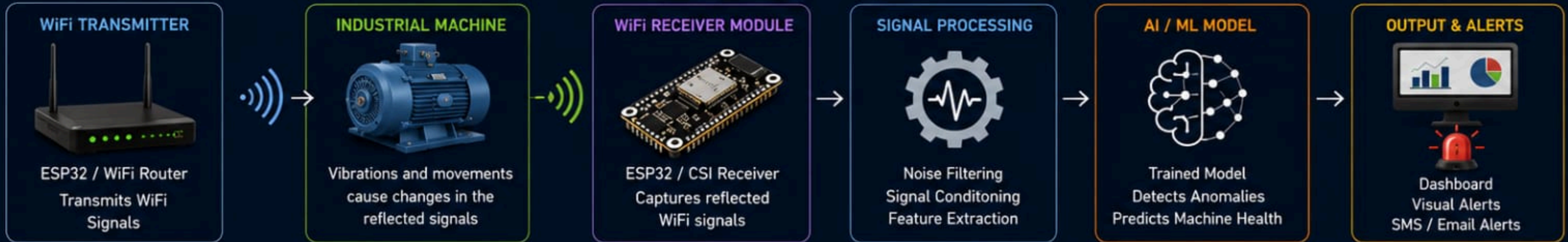


Strong market
demand

FACTORYSENSE

CONTACTLESS MACHINE HEALTH MONITORING SYSTEM USING WiFi RADAR

SYSTEM BLOCK DIAGRAM



WORKING FLOW



HARDWARE COMPONENTS



DASHBOARD PREVIEW



KEY BENEFITS FOR MSMEs



FactorySense – Smarter Machines, Safer Workplaces, Stronger MSMEs.



Contactless



Real-time Monitoring



Predictive Maintenance



Industry 4.0 Ready



IMPLEMENTATION STRATEGY

FactorySense Prototype Development



FactorySense

Contactless Machine
Health Monitoring

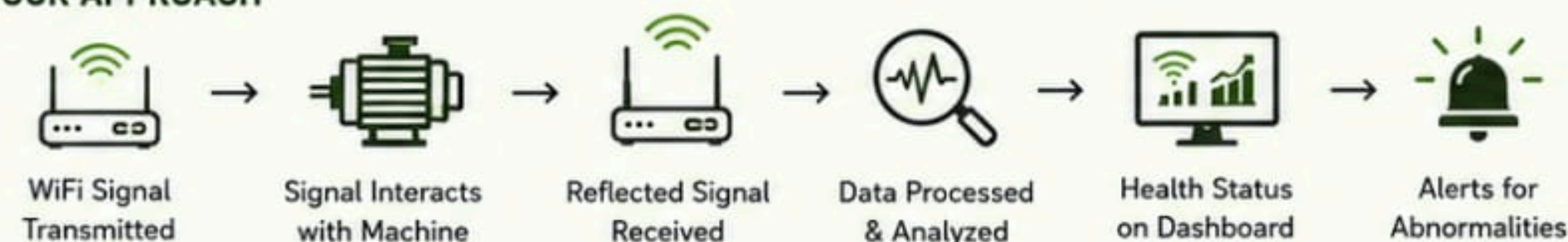


PHASE		IMPLEMENTATION
1		Hardware Setup Install one ESP32 as the WiFi transmitter and another ESP32 as the CSI receiver near the machine.
2		Data Collection Record WiFi signal patterns while the machine operates under normal conditions.
3		Signal Processing Filter the received CSI data to identify changes caused by machine vibrations.
4		Fault Detection Compare current readings with the normal pattern to detect abnormal vibration.
5		Alert System Display machine status on a simple dashboard and trigger an LED / Buzzer alert when needed.

PROTOTYPE COMPONENTS		
	ESP32 Development Board	2
	WiFi Router / Mobile Hotspot	1
	Breadboard & Jumper Wires	1 Set
	LED + Buzzer	1
	Laptop (Dashboard)	1

ESTIMATED PROTOTYPE COST	
Item	Cost (₹)
2 × ESP32 Boards	900
Breadboard & Wires	250
LED & Buzzer	150
Power Supply	300
Miscellaneous	400
Total Prototype Cost	≈ ₹2,000

OUR APPROACH



Low-cost

Built using affordable components



Easy to Build

Simple setup with readily available parts



Easy to Demonstrate

Real-time results on dashboard



Scalable for Industries

Can be expanded to monitor multiple machines





Thank you so much